4.1: Organic chemistry is key to the origin of life

For historical reasons, compounds containing carbon are said to be organic, and their study is called **organic chemistry**. Organic compounds range from simple molecules, such as methane (CH_4), to colossal ones, such as proteins, with thousands of atoms.

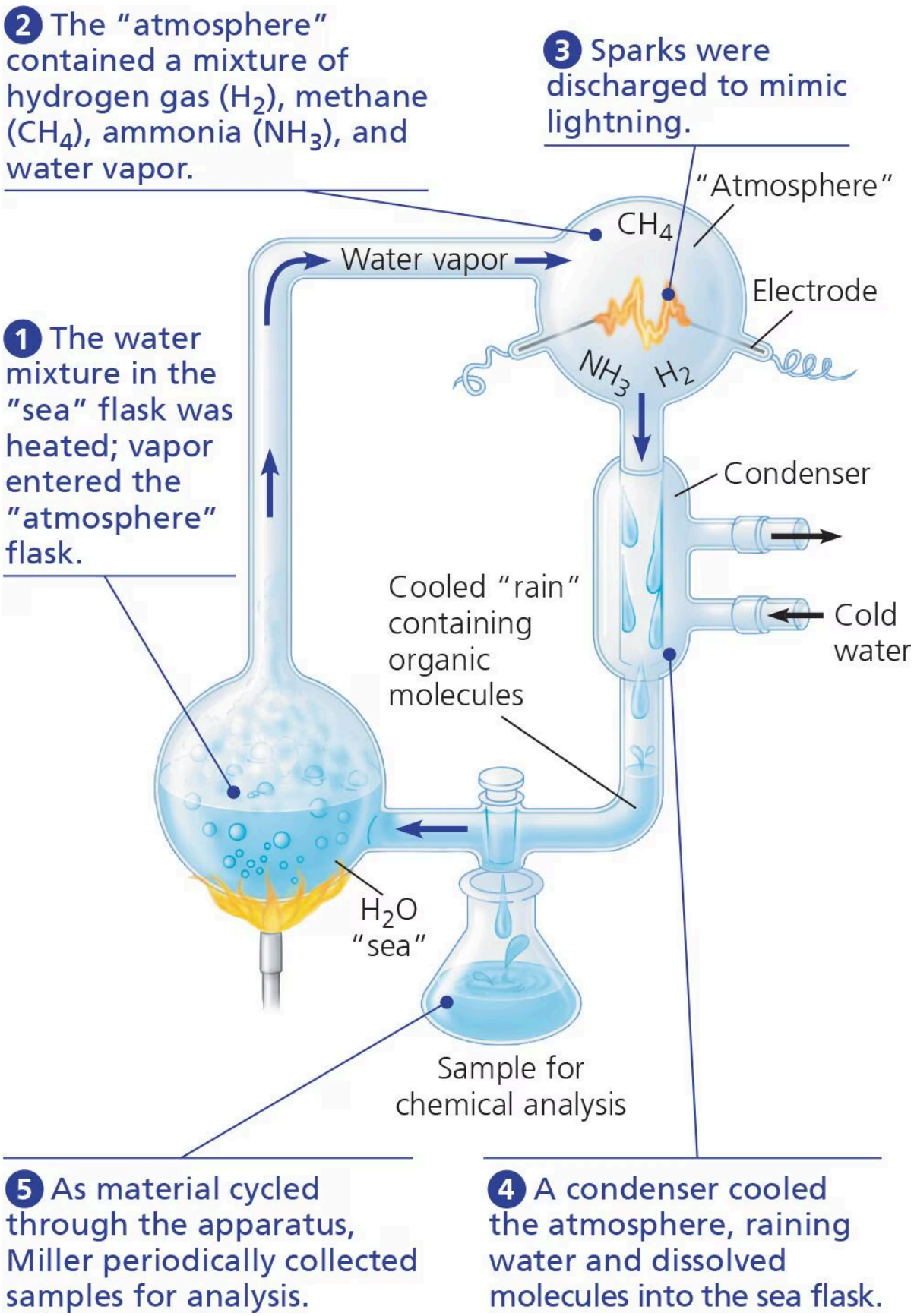
EVOLUTION In 1953, Stanley Miller, a graduate student of Harold Urey at the University of Chicago, designed an experiment on the abiotic (nonliving) synthesis of organic compounds to investigate the origin of life. Study **Figure 4.2** to learn about his classic experiment. From his results, Miller concluded that complex organic molecules could arise spontaneously under conditions thought at that time to have existed on early Earth. You can work with the data from a related experiment in the **Scientific Skills Exercise**. These experiments support the idea that abiotic synthesis of organic compounds, perhaps near volcanoes, could have been an early stage in the origin of life (see Figure 25.2).

Figure 4.2

Inquiry: Can organic molecules form under conditions estimated to simulate those on the early Earth?

Experiment

In 1953, Stanley Miller set up a closed system to mimic conditions thought at that time to have existed on the early Earth. A flask of water simulated the primeval sea. The water was heated so that some vaporized and moved into a second, higher flask containing the “atmosphere”—a mixture of gases. Sparks were discharged in the synthetic atmosphere to mimic lightning.


2 The "atmosphere" contained a mixture of hydrogen gas (H_2), methane (CH_4), ammonia (NH_3), and water vapor.
3 Sparks were discharged to mimic lightning.
"Atmosphere"
CH_4
Water vapor
Electrode
1 The water mixture in the "sea" flask was heated; vapor entered the "atmosphere" flask.
NH_3
H_2
Condenser
Cooled "rain" containing organic molecules
Cold water
H_2O "sea"
Sample for chemical analysis
5 As material cycled through the apparatus, Miller periodically collected samples for analysis.
4 A condenser cooled the atmosphere, raining water and dissolved molecules into the sea flask.

Experiment: in 1953, Stanley Miller set up a closed system to mimic conditions thought at the time to have existed on early Earth. A flask of water simulated the primeval sea. The water was heated so that some vaporized and moved into a second, higher flask containing the atmosphere- a mixture of gasses. Sparks were discharged by electrodes in the synthetic atmosphere to simulate lightning. Pictured is the experimental apparatus consisting of flasks connected by tubes that allow for the movement of water as vapor when heated or, when cooled, organic molecule containing rain, as well as flasks that allow for the collection of samples for chemical analysis.

Results

Miller identified a variety of organic molecules that are common in organisms. These included simple compounds, such as formaldehyde (CH_2O) and hydrogen cyanide (HCN), and more complex molecules, such as amino acids and long chains of carbon and hydrogen known as hydrocarbons.

Conclusion

Organic molecules, a first step in the origin of life, may have been synthesized abiotically on the early Earth. Although later evidence indicated that the early-Earth atmosphere was different from the “atmosphere” used by Miller in this experiment, recent experiments using the revised list of chemicals also produced organic molecules. (We will explore this hypothesis in more detail in Concept 25.1.)

Data from S. L. Miller, A production of amino acids under possible primitive Earth conditions, *Science* 117:528–529 (1953).

WHAT IF? *If Miller had increased the concentration of* NH_3 *in his experiment, how might the relative amounts of the products* HCN *and* CH_2O *have differed?*

*For suggested answer, see*Appendix A*.*

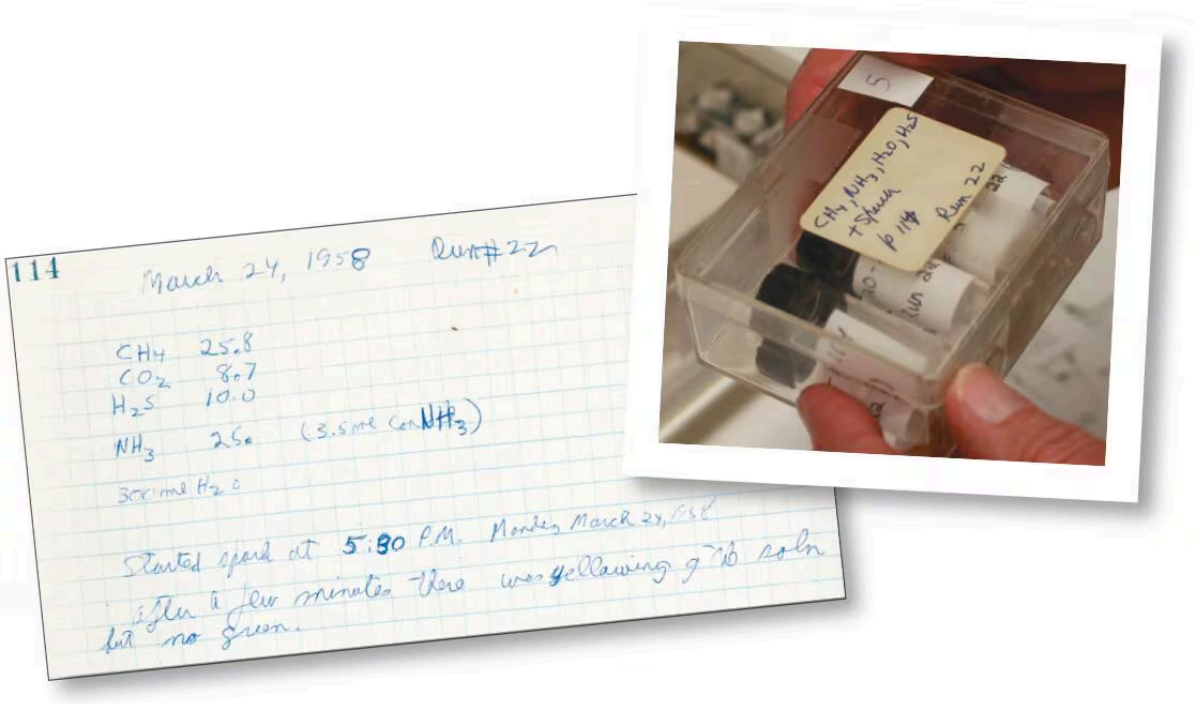
Scientific Skills Exercise

Working with Moles and Molar Ratios

AP Science Practices 5, 6

Could the First Biological Molecules Have Formed Near Volcanoes on Early Earth? In 2007, Jeffrey Bada, a former graduate student of Stanley Miller, discovered some vials of samples that had never been analyzed from an experiment performed by Miller in 1958. In that experiment, Miller used hydrogen sulfide gas (H_2S) as one of the gases in the reactant mixture. Since H_2S is released by volcanoes, the H_2S

experiment was designed to mimic conditions near volcanoes on early Earth. In 2011, Bada and colleagues published the results of their analysis of these “lost” samples. In this exercise, you will make calculations using the molar ratios of reactants and products from the H2S experiment.



A note from Stanley Miller's journal numbered 114 on the top left and dated March 24, 1958 is shown. An experimental run number 22 is also seen. The following is written below:

C H 4: 25.8

C O 2: 8.7

H 2 S: 10.0

N H 3: 25. (3.5 m L con N H 3)

300 ml H 2 O

Started spark at 5:30 p m. Monday March 24, 1958

after a few minutes there was yellowing of the solution but no green

A photograph of a box of his vials labeled as C H 4, N H 3, H 2 O, H 2 S plus spark p 114 Run 22 is shown.

Some of Stanley Miller’s notes from his 1958 hydrogen sulfide (H2S) experiment along with his original vials.

How the Experiment Was Done According to his laboratory notebook, Miller used the same apparatus as in his original experiment (see Figure 4.2), but the mixture of gaseous reactants included methane (CH_4), carbon dioxide (CO_2), hydrogen sulfide (H_2S), and ammonia (NH_3). After three days of simulated volcanic activity, he collected samples of the liquid, partially purified the chemicals, and sealed the samples in sterile vials. In 2011, Bada's research team used modern analytical methods to analyze the products in the vials for the presence of amino acids, the building blocks of proteins.

Data from the Experiment The table below shows 4 of the 23 amino acids detected in the 2011 analysis of the samples from Miller's 1958 H_2S experiment.

Product Compound	Molecular Formula	Molar Ratio (Relative to Glycine)
Glycine	$C_2H_5NO_2$	1.0
Serine	$C_3H_7NO_3$	3.0×10^{-2}
Methionine	$C_5H_{11}NO_2S$	1.8×10^{-3}
Alanine	$C_3H_7NO_2$	1.1

Product compound	Molecule formula	Molar ratio (relative to Glycine)
Glycine	C 2 H 5 N O 2	1
Serine	C 3 H 7 N O 3	3.0 times 10 raised to the negative second power
Methionine	C 5 H 11 N O 2 S	1.8 times 10 raised to the negative third power
Alanine	C 3 H 7 N O 2	1.1

Data from E. T. Parker et al., Primordial synthesis of amines and amino acids in a 1958 Miller H_2S-rich spark discharge experiment, *Proceedings of the National Academy of Sciences USA* 108:5526-5531 (2011). www.pnas.org/cgi/doi/10.1073/pnas.1019191108.

INTERPRET THE DATA

- A *mole* is the number of particles of a substance with a mass equivalent to its molecular (or atomic) mass in daltons. There are 6.02×1023molecules (or atoms) in 1.0 mole (Avogadro's number; see Concept 3.2). The data table shows the "molar ratios" of some of the products from the Miller H_2S experiment. In a molar ratio, each unitless value is expressed relative to a standard for that

experiment. Here, the standard is the number of moles of the amino acid glycine, which is set to a value of 1.0. For instance, serine has a molar ratio of 3.0×10^{-2}, meaning that for every mole of glycine, there is 3.0×10^{-2} mole of serine.

- Give the molar ratio of methionine to glycine and explain what it means.
- How many molecules of glycine are present in 1.0 mole?
- For every 1.0 mole of glycine in the sample, how many molecules of methionine are present? (Recall that to multiply two numbers with exponents, you add their exponents; to divide them, you subtract the exponent in the denominator from that in the numerator.)
- ◦ Which amino acid is present in higher amounts than glycine?
- How many more molecules of that amino acid are present than the number of molecules in 1.0 mole of glycine?
- The synthesis of products is limited by the amount of reactants.
- If one mole each of CH_4, NH_3, H_2S, and CO_2 is added to 1 liter of water (=55.5 moles of H_2O) in a flask, how many moles of hydrogen, carbon, oxygen, nitrogen, and sulfur are in the flask?
- Looking at the molecular formula in the table, how many moles of each element would be needed to make 1.0 mole of glycine?
- What is the maximum number of moles of glycine that could be made in that flask, with the specified ingredients, if no other molecules were made? Explain.
- If serine or methionine were made individually, which element(s) would be used up first for each? How much of each product could be made?
- The earlier published experiment carried out by Miller did not include H_2S in the reactants (see Figure 4.2). Which of the compounds shown in the data table can be made in the H_2S experiment but could not be made in the earlier experiment?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

In Concept 3.2, you learned about evidence for the presence of water on Mars. Even more exciting, in 2018, NASA reported that the rover *Curiosity* had found carbon-based compounds on Mars in a crater where a lake once existed. While these compounds might have been brought to Mars on a meteorite or formed by geologic processes, an intriguing possibility is that they might have been the relics of life-forms that once existed on that planet.

The overall percentages of the major elements of life—C, H, O, N, S, and P—are quite uniform from one organism to another, reflecting the common evolutionary origin of all life. Because of carbon's ability to form four bonds, however, this limited assortment of atomic building blocks can be used to build an inexhaustible variety of organic molecules. Different species of organisms, and different individuals within a species, are distinguished by variations in the types of organic molecules they make. In a sense, the great diversity of living organisms we see on the planet (and in fossil remains) is made possible by the unique chemical versatility of the carbon atom.

Interview

Interview with Stanley Miller: Investigating the origin of life